D06 · Cold Chain & Specialized Logistics

02 · Hazmat logistics: ADR/DOT compliance & dangerous goods

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1 WHAT IT IS

Hazmat logistics is the regulated movement of dangerous goods: correct UN classification, approved packaging and labeling, required documentation, qualified carriers and trained personnel, under national rules (NOM-002-SCT and related in Mexico, DOT in the US) and international codes (ADR, IMDG, IATA DGR).

2 WHY IT MATTERS

Dangerous goods errors are not service failures; they are safety and legal failures. **Responsibility sits with the shipper as well as the carrier** — misclassification, wrong packaging or an untrained person preparing the shipment can lead to rejection, fines, accidents and legal liability. The process must be designed so that a non-compliant shipment cannot be released.

3 HOW IT IS DONE

1. **Keep one controlled classification master.** UN number, class, packing group and proper shipping name per SKU, owned and reviewed.
2. **Enforce documentation in the system.** No release without a complete dangerous goods declaration and emergency information.
3. **Qualify carriers by hazard class and mode.** Authorization for general freight is not authorization for every class.
4. **Train and authorize people.** Initial and recurrent training, with an authorization matrix of who may prepare or offer DG shipments.

IN PRACTICE

Maintain a single, controlled source of truth for classifications

Multiple spreadsheets with divergent UN numbers are a common and dangerous failure mode.

Treat training as a formal authorization, not a one-time event

Only trained and currently certified personnel should be allowed to prepare or offer dangerous goods shipments.

Qualify carriers specifically for the hazard classes you ship

General freight authority does not equal dangerous goods authority for every class.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — Chemical distributor with recurring documentation rejections — Approximately 4% of hazmat shipments were delayed or rejected at origin or destination due to documentation or labeling errors.

Root Cause	Share of Rejections	Corrective Action
Incorrect or outdated UN classification	31%	Centralized classification master + annual review
Missing or incomplete DG Declaration	27%	System-enforced document checklist before release
Labeling / marking errors	22%	Standardized label kits + photo verification

Root Cause	Share of Rejections	Corrective Action
Untrained personnel preparing shipments	20%	Mandatory recurrent training + authorization matrix

Result: Rejection rate fell from 4% to 0.6% within nine months. More importantly, the company passed a subsequent SCT-related inspection without major findings.

HOW IT IS MEASURED

Hazmat Documentation Accuracy %

(Shipments with complete and correct DG documentation / Total hazmat shipments) × 100

Benchmark: >=99.5% · anything lower indicates systemic process failure

Training Compliance %

(Personnel with current required DG training / Personnel who handle DG) × 100

Benchmark: 100% · expired training is a regulatory violation

WHAT YOU WOULD USE

Dangerous Goods Compliance

Labelmaster / specialized dangerous-goods software — Specialized dangerous goods software for classification, documentation and labeling.

TMS modules with DG checks — Leading TMS platforms can enforce documentation and carrier qualification rules before shipment release.

4 SO WHAT

Make non-compliance impossible to ship. One classification master, system-enforced documents, carriers qualified per class, and trained, authorized people.

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